PORTFOLIO IQ
EQUITY RISK AND RETURNS CHALLENGE

TEAM NAME : BRB

TEAM NO:9

MEMBERS NAME : ASHMIT SHUKLA ,SHASHWAT SINGH ,SURYANSH SINGH, AYUSH BAJPAI

TASK 1 : DATA ACQUISITION AND QUALITY

❖ Dataset Overview

Source: Yahoo Finance (yfinance)

Universe: 15 NSE-listed stocks + Nifty 50 benchmark

Time Period: Jan 2023 – Dec 2024

Trading Days per Stock: 491

Total Records: 7,856 rows and 9 columns

❖ Data Quality Metrics

Missing Values: 0

Duplicate Rows:0

Duplicate (Ticker, Date):0

Invalid OHLC Entries:0

Negative Volume:0

Date Alignment with Nifty 50: Perfect

❖ Final Status

Dataset Quality:
Clean and Complete

Ready for Analysis:
Yes

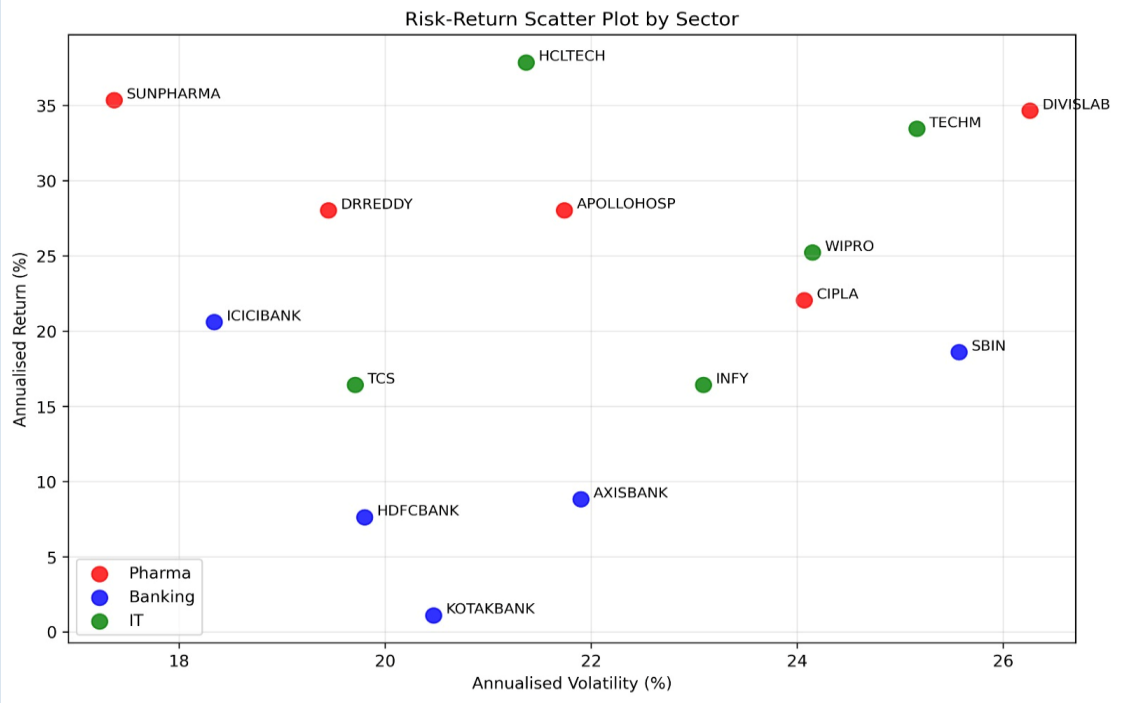
Imputation Required:
No

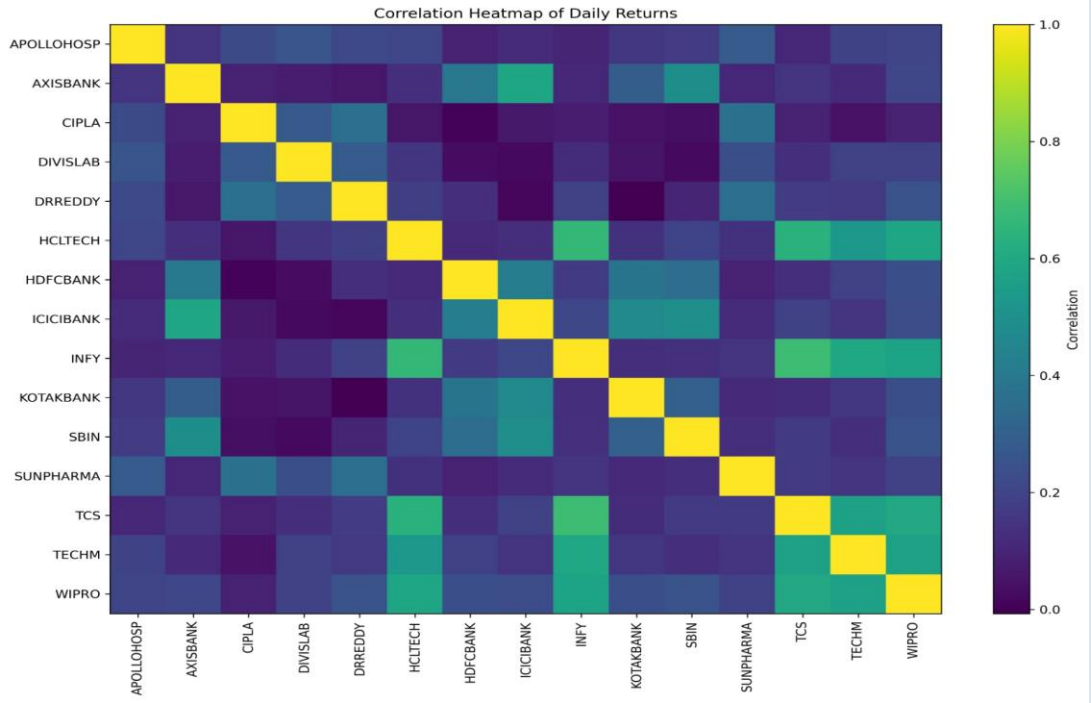
TASK 2 : RISK AND RETURN ANALYSIS

SUMMARY TABLE

Ticker	Annualised Return (%)	Annualised Volatility (%)	Sharpe Ratio	Beta vs NIFTY50	Maximum Drawdown (%)	SMA_50	SMA_200
APOLLOHOSP	28.03	21.74	0.99	0.564	-14.87	7056.82	6565.78
AXISBANK	8.81	21.9	0.105	1.06	-19.18	1143.2	1160.5
CIPLA	22.04	24.07	0.646	0.332	-20.77	1486.84	1493.14
DIVISLAB	34.65	26.26	1.072	0.453	-21.45	5902.24	4851.42
DRREDDY	28.03	19.45	1.107	0.429	-15.61	1264.83	1268.54
HCLTECH	37.86	21.37	1.468	0.787	-21.66	1793.46	1560.2
HDFCBANK	7.62	19.8	0.057	1.039	-19.91	877.13	805.41
ICICIBANK	20.59	18.34	0.768	0.97	-9.45	1283.17	1189.13
INFY	16.42	23.09	0.43	0.902	-24.34	1824.44	1650.15
KOTAKBANK	1.1	20.47	-0.264	0.863	-23.15	351.79	353.39
SBIN	18.59	25.57	0.473	1.391	-17.48	811.96	795.25
SUNPHARMA	35.34	17.37	1.66	0.473	-12.24	1789.58	1662.53
TCS	16.41	19.71	0.503	0.747	-13.17	3988.15	3898.83
TECHM	33.44	25.16	1.071	0.873	-16.22	1666.05	1441.97

SCATTERPLOTS AND HEATMAP VISUALISATION





KEY INSIGHTS

- **SUNPHARMA appears mispriced relative to its pharma sector peers on the Risk-Return chart. It delivered one of the highest annualized returns in the universe at 35.34%, while also maintaining one of the lowest annualized volatilities at 17.37%, which is notably better than other pharma names such as CIPLA, DIVISLAB, and APOLLOHOSP. This combination of high return and relatively low risk makes SUNPHARMA stand out as offering stronger risk-adjusted performance than its sector peers.**

- **The two most correlated stocks are TCS and INFY, with a correlation of about 0.687 in daily returns. Holding both in the same portfolio increases concentration risk because they are likely to react similarly to the same sector-wide drivers, reducing the diversification benefit. If the IT sector faces a negative shock, both positions may decline together rather than offsetting each other.**

TOP 3 STOCKS WITH HIGHEST SHARPE RATIO

Ticker	Annualised Return (%)	Annualised Volatility (%)	Sharpe Ratio	Beta vs NIFTY50	Maximum Drawdown (%)	SMA_50	SMA_200
SUNPHARMA	35.34	17.37	1.66	0.473	-12.24	1789.58	1662.53
HCLTECH	37.86	21.37	1.468	0.787	-21.66	1793.46	1560.2
DRREDDY	28.03	19.45	1.107	0.429	-15.61	1264.83	1268.54

- **SUNPHARMA:** Consistent returns with stable price movement
- **HCLTECH:** High returns with efficient risk utilization
- **DRREDDY:** Balanced performance with moderate volatility

LAST 3 STOCKS WITH LOWEST SHARPE RATIO

Ticker	Annualised Return (%)	Annualised Volatility (%)	Sharpe Ratio	Beta vs NIFTY50	Maximum Drawdown (%)	SMA_50	SMA_200
AXISBANK	8.81	21.9	0.105	1.06	-19.18	1143.2	1160.5
HDFCBANK	7.62	19.8	0.057	1.039	-19.91	877.13	805.41
KOTAKBANK	1.1	20.47	-0.264	0.863	-23.15	351.79	353.39

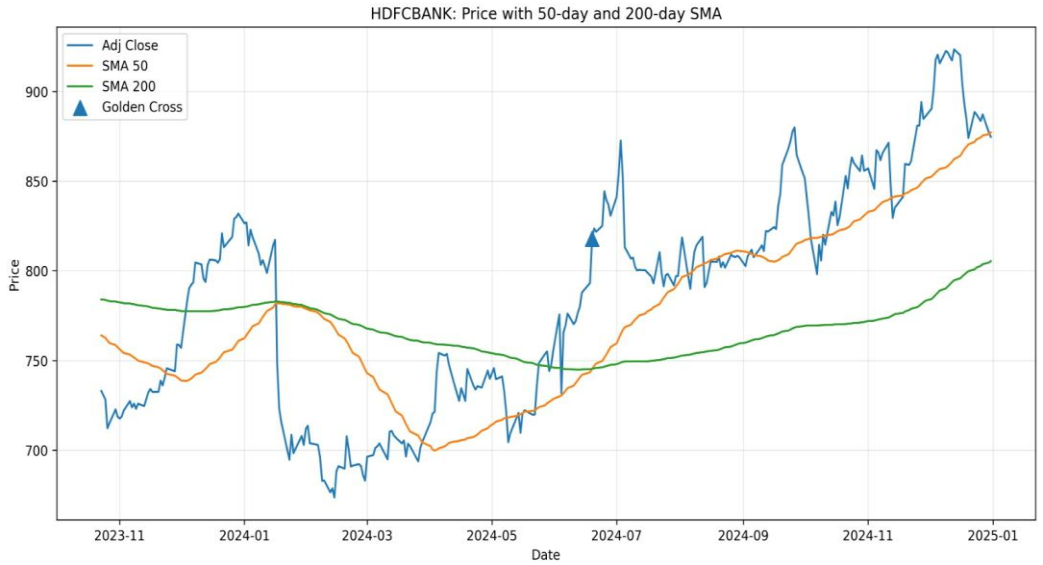
- **AXISBANK:** Moderate returns but relatively higher volatility
- **HDFCBANK:** Stable performance but lower return generation
- **KOTAKBANK:** Weak returns compared to risk taken

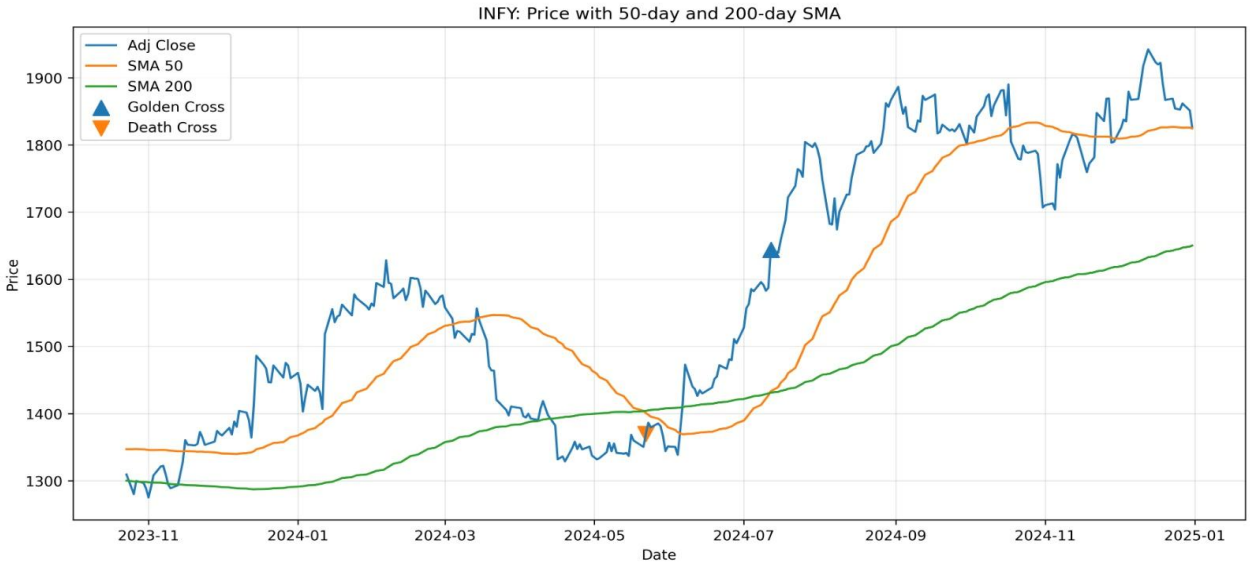
TASK 3 :TECHNICAL SIGNAL DASHBOARD

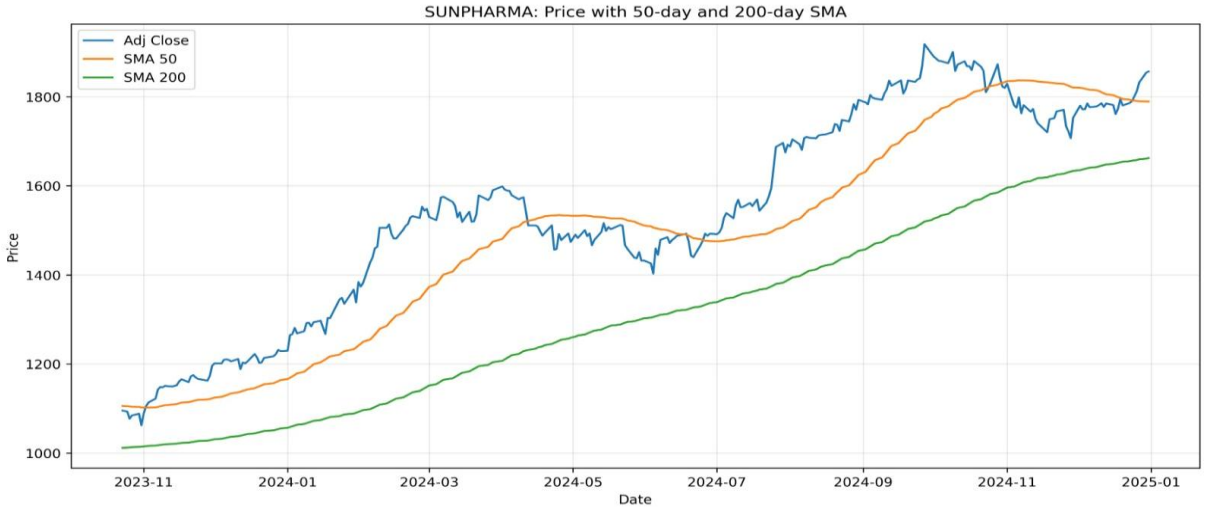
SIGNAL TABLE

Ticker	SMA_50	SMA_200	Signal	Date of Last Crossover	Last Crossover Event
APOLLOHOSP	7056.82	6565.78	Golden Cross	No crossover in period	No crossover in period
AXISBANK	1143.2	1160.5	Death Cross	2024-12-12 00:00:00	Death Cross
CIPLA	1486.84	1493.14	Death Cross	2024-12-24 00:00:00	Death Cross
DIVISLAB	5902.24	4851.42	Golden Cross	2024-05-16 00:00:00	Golden Cross
DRREDDY	1264.83	1268.54	Death Cross	2024-12-13 00:00:00	Death Cross
HCLTECH	1793.46	1560.2	Golden Cross	2024-07-22 00:00:00	Golden Cross
HDFCBANK	877.13	805.41	Golden Cross	2024-06-19 00:00:00	Golden Cross
ICICIBANK	1283.17	1189.13	Golden Cross	No crossover in period	No crossover in period
INFY	1824.44	1650.15	Golden Cross	2024-07-12 00:00:00	Golden Cross
KOTAKBANK	351.79	353.39	Death Cross	2024-12-26 00:00:00	Death Cross
SBIN	811.96	795.25	Golden Cross	No crossover in period	No crossover in period
SUNPHARMA	1789.58	1662.53	Golden Cross	No crossover in period	No crossover in period
TCS	3988.15	3898.83	Golden Cross	No crossover in period	No crossover in period
TECHM	1666.05	1441.97	Golden Cross	No crossover in period	No crossover in period
WIPRO	272.5	244.27	Golden Cross	No crossover in period	No crossover in period

TASK 3 :TECHNICAL SIGNAL DASHBOARD







TASK 3 :TECHNICAL SIGNAL DASHBOARD

INFY is a good example of a Golden Cross that worked. The Golden Cross occurred on 12 July 2024, when the 50-day SMA moved above the 200-day SMA. After that signal, the stock rose from about 1643.83 on the crossover date to about 1726.08 within roughly a month, and later closed the period near 1825.96. This suggests the crossover was followed by genuine upward momentum rather than an immediate reversal. At the same time, because the signal came after part of the recovery had already started, it shows that SMA crossovers are lagging indicators and should be used as confirmation tools rather than in isolation.

TASK 4 : PORTFOLIO CONSTRUCTION

PORTFOLIO COMPARISON

Portfolio	Annualised Return (%)	Annualised Volatility (%)	Sharpe Ratio	Banking Exposure (%)	IT Exposure (%)	Pharma Exposure (%)
Portfolio A	22.28	11.42	1.382	33.33	33.33	33.33
Portfolio B	28.2	11.65	1.863	18	34	48

SECTOR BREAKDOWN COMPARISION

Sector	Average_Sharpe_Ratio	Average_Beta_vs_NIFTY50
Banking	0.228	1.065
IT	0.849	0.876
Pharma	1.095	0.45

SETOR EXPOSURE BREAKDOWN

Sector	Portfolio A Exposure (%)	Portfolio B Exposure (%)
Banking	33.33333333	18
IT	33.33333333	34
Pharma	33.33333333	48

TASK 4 :PORTFOLIO CONSTRUCTION

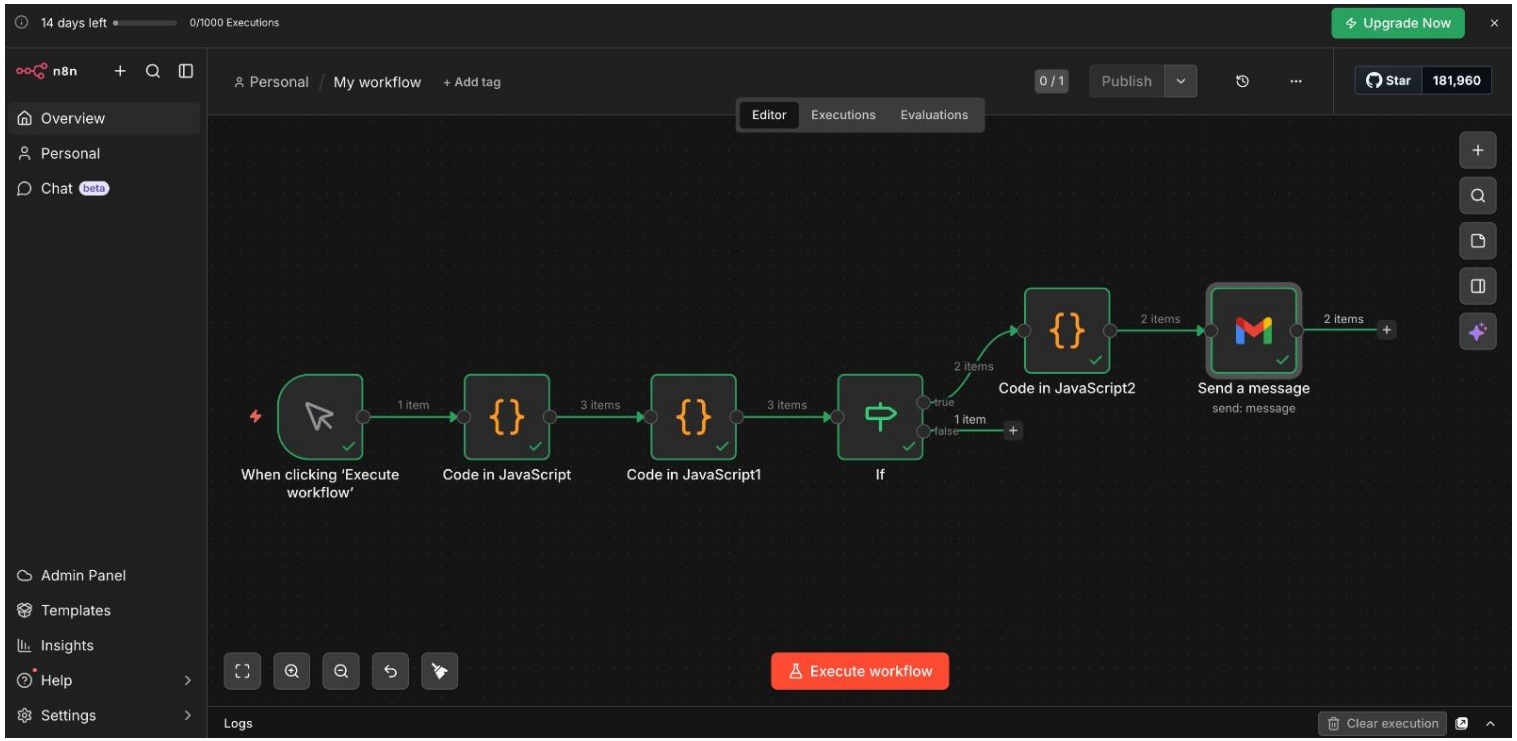
Portfolio B is a selective, high-conviction portfolio designed from the stock-level risk and return analysis rather than a mechanical equal-weight approach. We overweighted SUNPHARMA, HCLTECH, DRREDDY, DIVISLAB, APOLLOHOSP and ICICIBANK because these names offered the strongest combination of return potential, Sharpe Ratio and manageable downside. SUNPHARMA was given the highest weight because it combined high annualized return with the best Sharpe Ratio and relatively low beta. HCLTECH and TECHM strengthened the IT allocation due to strong risk-adjusted performance, while DRREDDY, DIVISLAB and APOLLOHOSP supported the portfolio's pharma tilt, which was justified by the sector's superior average Sharpe Ratio and lower beta.

We excluded KOTAKBANK because it had the weakest return profile and a negative Sharpe Ratio, making it unattractive on a risk-adjusted basis. We also underweighted HDFCBANK and AXISBANK because banking, as a sector, showed weaker Sharpe performance and higher market sensitivity than Pharma and IT. ICICIBANK was retained as the core banking position because of its relatively low drawdown and balanced profile.

The portfolio has a moderately aggressive risk profile: it aims for higher return without a proportionate rise in volatility. This is evident in Portfolio B's stronger annualized return of 28.20% and Sharpe Ratio of 1.863 versus Portfolio A's 22.28% and 1.382. It suits an investor seeking better risk-adjusted growth with controlled risk.

TASK 5: AUTOMATION VIA N8N

SCREENSHOT OF N8N WORKFLOW



LINK OF WORKFLOW JSON:
https://drive.google.com/file/d/11Wn6kGDhZmeOe8gb55Sy20U0bnDy2LSt/view?usp=drive_link

TASK 6: CHAOS ROUND

	Portfolio	Portfolio Beta	Expected Loss if NIFTY50 drops 10% (%)
0	Portfolio A	0.797	-7.97
1	Portfolio B	0.712	-7.12

Portfolio B is better positioned for the Chaos Round because it combines lower market sensitivity with stronger historical efficiency. Its beta of 0.712 implies a smaller expected loss than Portfolio A's 0.797 in a 10% Nifty decline. This is consistent with the portfolio's construction: Banking exposure was cut from 33.33% to 18.00% because Banking had the weakest average Sharpe Ratio and the highest beta, while Pharma exposure was increased from 33.33% to 48.00% because Pharma had the strongest Sharpe profile and the lowest beta. As a result, Portfolio B is both more defensive in a stress event and superior on risk-adjusted performance.

TASK 6: CHAOS ROUND

MOST EXPOSED AND SAFEST STOCKS

While CIPLA is the safest stock by pure beta, SUNPHARMA is the strongest defensive refuge overall. Its beta of 0.473 implies a relatively limited expected loss of 4.73% in the stress scenario, but unlike CIPLA it also offers the best Sharpe Ratio in the universe at 1.660, a high annualized return of 35.34%, and a relatively shallow maximum drawdown of 12.24%. This makes SUNPHARMA the more convincing safe-refuge choice when both downside resilience and quality of returns are considered.

By pure beta math, CIPLA is the safest refuge because it has the lowest beta and lowest expected crash loss. But from a broader portfolio perspective, SUNPHARMA is the better defensive holding because it combines low beta with the best Sharpe Ratio, strong returns, and limited drawdown.

TASK 6: CHAOS ROUND

MOST EXPOSED AND SAFEST STOCKS

SBIN is the most exposed stock because it has the highest beta in the universe at 1.391, implying an expected loss of 13.91% in a 10% Nifty crash. This is supported by its relatively high annualized volatility of 25.57% and only moderate Sharpe Ratio of 0.473, which suggests that it carries substantial market risk without delivering the strongest risk-adjusted return.

Thank you